

Temporal counterpart theory

Advanced Intro to Metaphysics

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1. Bare bones counterpart theory

Basic Counterpart Theory for an arbitrary modal operator $\diamond$: there is a relation, $\diamond$ -*counterparthood*, such that for any qualitative property, to $\diamond$ have that property is to have a $\diamond$ -counterpart that has it.

We can say this in higher-order logic

$$\text{BCT: } \exists C \forall F (\text{Qual}(F) \rightarrow (\lambda x. \diamond Fx) = (\lambda x. \exists y (Cyx \wedge Fy)))$$

Here the variables C, F, x, y are of types $\langle e, e \rangle, \langle \langle e \rangle \rangle, e, e$.

- BCT is inspired by some informal expositions of counterpart theory, e.g. according to Sider ('Beyond the Humphrey Objection'), 'David Lewis's modal counterpart theory identifies possibly being F with having a counterpart... that is F '. Such formulations typically do not include the restriction to qualitative F . However the stronger version of BCT that drops this restriction is subject to some major objections that BCT avoids.
- BCT is relative to a choice of interpretation of $\diamond$; it could be true for some $\diamond$ and false for others. For metaphysical possibility, it's vulnerable to the 'incredulous stare' objection that *possibly riding a talking donkey* is qualitative and instantiated, although *having a counterpart that rides a talking donkey* isn't (since there aren't any talking donkeys). That doesn't automatically undermine the proposal that BCT holds for some other $\diamond$.
- There's no mention in BCT of 'world', 'in', 'actual'; no claim about the role of similarity; no special operation of restricting quantifiers. That's not to say that BCT is *inconsistent* with further claims that might mention such things.
- There's nothing to force any revisionism in either nonmodal or modal logic—and I think the best way of developing the view avoids such revisionism! It does imply necessitism though. Other logical properties of $\diamond$ will impose constraints on the witnessing C —e.g. Iteration suggests that it should be transitive.
- Even if we settle on a particular C as the witness to BCT, the result clearly falls short of a 'general theory of $\diamond$ '. For example it provides no nontrivial answer to the question $(\lambda xy. \diamond Rxy) = ?$, even for qualitative R .

2. Bare bones temporal counterpart theory

Basic Temporal Counterpart Theory, BTCT, applies BCT to the future tense operator F and the past tense operator P . Let's call the witnesses *future-counterparthood*, C_F and

past-counterparthood, C_P , respectively. (Don't worry if there are multiple witnesses.)

- It's plausible that future-counterparthood and past-counterparthood are mutual converses.
- BTCT has some incredulous-stare provoking consequences of its own. Since uncountably many real numbers have been my age in years while I was a grad student, and *being a grad student who is x years old* is qualitative for each real number x , it implies that there are uncountably many grad students!
- *Pace Sider*, BTCT does not imply that there are uncountably many *North American* grad students, since *being North American* is not qualitative. Better to say that past-counterparts of me live not in Europe and North America but in their past-counterparts.
- A contrast with the modal case: the existence of a four-dimensional manifold of spacetime points is independently motivated. The BTCTist needs no new or controversial posits as regards *fundamental* ontology; the required abundance of people (etc.) can be explained via a nonstandard account of what it is to be a person (etc.).
- Sider's view: people just are certain 3d spacetime regions. (Or at least related to such regions by a one-one 'exact occupation' relation.) This goes with "strong monism" in Fine's terminology. A Plenitudinous ontology generates an even wider range of candidates to be the people, and makes it easy to be a pluralist.

3. The eternity of the qualitative

Suppose BTCT is true and p is a qualitative proposition.

Then $\lambda x.p$ is a qualitative property. So we can instantiate to get

$$(\lambda x. F(\lambda x.p)x) = (\lambda x. \exists y (C_F yx \wedge (\lambda x.p)y)).$$

Using Leibniz's Law, we apply both sides to some given a to get

$$(\lambda x. F(\lambda x.p)x)a = (\lambda x. \exists y (C_F yx \wedge (\lambda x.p)y)a$$

Given the principle of *Beta-conversion*, this simplifies to

$$Fp = \exists y (C_F ya \wedge p)$$

which implies $Fp \rightarrow p$. Since p was an arbitrary qualitative proposition, we have $\forall p(\text{Qual}(p) \rightarrow (Fp \rightarrow p))$. Anything qualitative that will ever be the case is (already) the case.

4. Counterparts of properties and propositions

It's natural to want to talk too of properties, relations, and propositions as counterparts. We can keep BTCT as it stands, just allowing the variables x and y to be of some type other than e .

To get this to harmonize with BTCT for objects, we will need tight connections between counterparthood in different types. For example: let

a: David Lewis

F: owning a train set

G: being owned by David Lewis

H: being instantiated by a train set.

Then we want $Fa = HG$, or at least the two should be modally interchangeable. Since F and H are qualitative, the natural way to achieve this is to say that for any property X, X is a counterpart of G iff for some x , x is a counterpart of David Lewis and X is *being owned by x*.

5. Worlds and times

Standard assumption: instants of time are certain slices through spacetime (3d regions).

But in BTCT, these slices do not play anything like the role of instants of time.

- This role involves principles like 'It will be the case that p iff at some future time p '. Hard to think of a sensible interpretation of 'at' and 'future' that could make this compatible with BTCT and the identification of times with slices.

A more promising idea: instants of time are certain *propositions*. Always exactly one of these propositions is true.

- 'At t , p ' means 'Sometimes $t \wedge p$ ' (i.e. $P(t \wedge p) \vee (t \wedge p) \vee F(t \wedge p)$).
- 'Future' means $\lambda p.Fp$. 'Present' means $\lambda p.p$.
- There are many collections of propositions which play this role. Which should we identify with the times? This involves questions about the "transworld identity" of times; it may turn out to be rather holistic, messy, vague, context-dependent...

6. Temporal counterparts of spacetime points